UNIT 3

1. Data Transformation:

Data Transformation is the process of converting raw data into a clean, structured, and analysis-ready format. In tools like Power BI, Power Query, Excel, or SQL, transformation ensures that the data is accurate, consistent, and useful for analysis or reporting.

Data Transformation is the process of modifying, cleaning, structuring, and enriching raw data into a suitable format for analysis and visualization.

Data Transformation is a key step in ETL (Extract → Transform → Load).

In Power BI, data transformation is mainly done using Power Query Editor, which allows users to perform ETL (Extract, Transform, Load) operations.

Data Transformation is Important

- Removes errors, duplicates, and inconsistencies.
- Converts data into a standard format.
- Improves data quality and reliability.
- Helps in building dashboards, reports, and predictive models.
- Ensures data follows business rules.

Steps in Data Transformation

a. Data Cleaning

- Handling missing values (replace/remove)
- Removing duplicates
- Correcting spelling errors
- Standardizing formats (date, numbers, text)

b. Data Structuring

- Adding new columns
- Splitting/merging columns
- Changing data types
- Creating calculated fields

c. Data Integration

- Combining data from multiple sources
- Merging or appending tables

d. Data Reduction

- Filtering rows
- Removing unwanted columns
- Aggregating (sum, average, group-by)

2. Power Query Editor:

Power Query is the data transformation engine inside Power BI. Power Query Editor is a data transformation and preparation tool available in **Power BI**, **Excel**, and **Power Query Online**.It allows users to **connect, clean, reshape, and prepare** data before loading it into the data model.Power Query works using a backend formula language called **M Language** (Mashup Language). Each transformation you apply becomes a **step** in the "Applied Steps" pane.

Features of Power Query

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- Import from multiple sources
- Automatically records steps
- User-friendly UI
- Supports formulas (M-language)
- Fast processing for large datasets

Power Query Editor is a **no-code**/low-code interface in Power BI Desktop where you:

- **Connect to data sources**
- **Clean and transform data**
- **Shape data before loading it into the data model**

To open it.

Home Tab → Transform Data → Power Query Editor

Use of Power Query Editor

- To clean and prepare data for analysis
- To automate repetitive cleaning tasks
- To merge data from multiple sources
- To maintain data accuracy and consistency
- To transform large datasets efficiently

Power BI dashboards heavily depend on correct transformations performed in Power Query.

Interface Overview

I. Ribbon Tabs

Power Query Editor contains several tabs:

a. Home Tab

Most commonly used operations:

- Remove columns/rows
- Keep rows
- Merge queries
- Append queries
- Use first row as header
- Close & apply

b. Transform Tab

Used to modify data WITHOUT adding new column:

- Data type
- Pivot / Unpivot columns
- Replace values
- Reverse rows
- Split columns
- Format text

c. Add Column Tab

Used to add new columns:

- Custom column
- Conditional column
- Index column

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- Duplicate column
- Extract values

d. View Tab

Used to control visibility:

- Formula bar
- Column quality
- Column distribution
- Advanced editor

Key Components of Power Query Editor Interface

I. Query Pane

Left side — shows all tables/queries you have loaded.

II. Data Preview Pane

Displays data after each transformation.

III. Applied Steps Pane

Right side — shows each transformation step.

You can:

- Rename steps
- Delete steps
- Reorder steps
- View M code for each step

Connecting and Importing Data

Power Query supports many data sources:

- Excel, CSV
- SQL Server, MySQL
- Web data
- SharePoint
- PDF
- Folder (for combining multiple files)

Steps

Home → New Source → Choose file → Open in Power Query Editor

Most Important Data Transformation Functions

Use First Row as Headers

Power Query uses "Column1, Column2" by default.

Steps

Home → *Use First Row as Headers*

Why?

Meaningful column names improve readability and accuracy.

Remove Rows

Used to eliminate unnecessary rows.

Options

- Remove Top Rows
- Remove Bottom Rows
- Remove Alternate Rows
- Remove Blank Rows
- Remove Duplicates

Usage

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Removing:

- Subtotals
- Summary lines
- Extra headers
- Blank or null rows

Keeping Rows
Opposite of removing rows.
Options
- Keep Top Rows
- Keep Bottom Rows
- Keep Range of Rows

Usage
Keep only required sample data.

Remove Columns
Used to drop irrelevant data.
Examples:
- Address fields not required
- Temporary columns
- Audit columns

Be careful: Removing columns early affects further steps.
Filter Rows
Filtering rows based on:
- Text
- Number
- Date
- Custom conditions

Examples:
- Age > 18
- State = "UP"
- Date between 2022–2023

Change Data Type
One of the most important transformation steps.
Common Types
- Whole Number
- Decimal Number
- Date
- Date/Time
- True/False
- Text

Incorrect data type results in:
- Errors
- Wrong visuals
- Wrong sorting

Split Column

Used when one column contains multiple values.

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Types

- By Delimiter (comma, space, '-')
- By Number of Characters
- By Positions

Examples

- Full Name → First Name, Last Name
- "123-ABC-789" → 3 parts

Merge Columns
Used to combine multiple fields.
Example:
FirstName + LastName = FullName
Replace Values
Used to correct data.
Examples:

- Replace "N/A" with null
- Replace "UP " with "UP"
- Replace misspelled category names

Fill Up / Fill Down
Used when data is missing but available in previous/next row.
Example:
Category listed once for multiple items.
Fill Down copies the above value; Fill Up copies below.
Group By
Used to create summarizations.
Examples

- Total Sales per Region
- Count Orders per Customer
- Average Marks per Student

Remove Errors
Fixes:

- Invalid data types
- Nulls causing errors
- Wrong calculations

Useful for cleaning CSV files.
Pivot & Unpivot Columns
a. Pivot Column
Converts rows into columns.
Example:
Month-wise Sales table.
b. Unpivot Column
Converts multiple columns into attribute-value pairs.
VERY important for making data analysis-friendly.
Example:
Year columns (2020, 2021, 2022) → Converted into:
Year | Value
Conditional Column

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Creates classification columns using IF-ELSE logic.

Example

If Score ≥ 90 → "Excellent", else "Good"

No need for formulas — UI based.

6.15 Custom Column

Used to write formulas using M Language.

Example

New column = [Quantity] * [Price]

Index Column

Creates Row Number:

- From 0
- From 1
- Custom

Useful for sorting or uniquely identifying rows.

Merge Queries

Used to combine columns from two tables.

Join Types

- Inner
- Left Outer
- Right Outer
- Full Outer
- Anti (Left/Right)

Example:

Add department name into student table using DeptID.

6.18 Append Queries

Used to STACK tables vertically.

Example

Sales_2023 + Sales_2024 = Combined Sales Table

3. **Remove Rows**

Used to clean data by removing irrelevant, duplicate, or unwanted rows.

Types of Remove Row Options

a) Remove Top Rows

- Deletes a specific number of rows from the top.
- Useful when:
  - Extra headings
  - Extra blank rows
  - File metadata is present

b) Remove Bottom Rows

- Removes the last few rows (common in system generated reports).

c) Remove Alternate Rows

- Used for repeated pattern data.

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- Example: Remove every 2nd or 3rd row.

d) Remove Blank Rows

- Automatically deletes fully blank rows.

e) Remove Duplicates

- Keeps only unique rows based on selected columns.

Steps

Home → **Remove Rows** → select option.

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4. **Add New Columns**

In **Power Query Editor**, adding new columns is essential for:

- Data transformation
- Data cleaning
- Feature engineering
- Preparing data for DAX & visualizations

The **Add Column** tab provides multiple ways to create new columns depending on your requirement.

1. Custom Column: A **Custom Column** is used to create formulas using Power Query's M-language.

- Creating calculated fields
- Mathematical operations
- Text manipulation
- Date calculations
- Logical operations

Example

Formula to calculate **GST = 18% of Sales**:

=[Sales] * 0.18

More Examples

a) Concatenate Text

=[FirstName] & " " & [LastName]

b) Check Even/Odd

= if Number.Mod([ID], 2) = 0 then "Even" else "Odd"

c) Calculate Profit

=[Revenue] - [Cost]

2. Index Column

Adds a **sequential number**:
0,1,2… or 1,2,3…

Steps

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Add Column → **Index Column**

Options

- From 0
- From 1
- Custom starting number

It Is Used

- To identify the **order of rows**
- To remove duplicates and keep the first/last row
- To create row numbers
- For sorting in visuals or modeling
- To group data and retain sequence

Example

Row	Name
1	A
2	B

3. Conditional Column

What It Is

Creates a new column based on IF…ELSE logic **without coding**.

Example

If Sales > 5000 → "High"
Else → "Low"

Best Use Cases

- Classification (High/Low, Yes/No, Pass/Fail)
- Flagging missing or incorrect values
- Creating categories (Age group, Risk level)
- Grouping sales or performance ranges

4. Duplicate Column

Creates a **copy of an existing column**.

When It Is Needed

- You want to transform a column but keep the original

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- Preparing multiple variations of same field
- Converting text → uppercase without losing original
- Splitting a column but keeping original

Steps

Add Column → **Duplicate Column**

Example

If you duplicate "Name" column and transform to uppercase:

Name	Name - Copy
Geeta	GEETA

6. Other Add Column Options

Power Query provides many built-in column types.

a) Column from Examples

- Type sample output → Power BI generates logic
- Great for: extracting substring, formatting text, combining data

Example:
Input: "ABC-2024"
Output: "2024"

Power Query learns pattern → extracts year.

b) Date & Time Columns

You can add new columns from dates:

- Year
- Month
- Quarter
- Day
- Week number
- Start of month
- End of month

Example:
OrderDate → extract Year:

2025-04-18 → 2025

c) Extract & Format Columns

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- Extract prefix/suffix
- Extract first N characters
- Format numeric or text fields

Steps to Add Any New Column

General Steps

1. Open **Power Query Editor**
2. Go to **Add Column** tab
3. Choose the type:
    - **Custom Column**
    - **Index Column**
    - **Conditional Column**
    - **Duplicate Column**
    - **Invoke Function**
    - **Column From Examples**
    - **Math / Date / Time columns**
4. Enter details → Click **OK**

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4-Conditional Column

A **Conditional Column** allows you to add a new column based on **logical rules** (IF…ELSE statements) **without writing M-code**.

It is one of the most commonly used tools in Power Query for data cleaning, data transformation, and classification.

1. How to Add a Conditional Column

Steps

1. Go to **Power Query Editor**
2. Click **Add Column** tab
3. Select **Conditional Column**

You will see a dialog box where you can define:

- Column Name
- Column to evaluate
- Condition (operator)
- Value
- Output
- Else output

2. Example Scenario (Basic IF Condition)

Condition

If Sales > 5000 → "High"
Else → "Low"

Steps

1. **Name of new column:** *Sales Category*
2. **Column Name:** Sales
3. **Operator:** is greater than
4. **Value:** 5000
5. **Output:** High
6. **Else:** Low

Result Table

Sales	Sales Category
7000	High
4000	Low

Why Conditional Column is Useful

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No coding needed

You can build IF statements visually.

Perfect for classification tasks

- Categorizing sales (High/Medium/Low)
- Grading students
- Risk levels (Low/High)
- Profit evaluation (Profit/Loss)

Works well for Data Cleaning

- Replace values based on condition
- Flag missing or incorrect entries
- Identify outliers

Helps in Feature Engineering

Multiple Conditions (Nested IF)

You can add multiple conditions by clicking **Add Clause**.

Example: Categorizing Sales

Rule:

- If Sales > 8000 → "Excellent"
- If Sales > 5000 → "Good"
- Else → "Average"

Setup in Conditional Column

1. Column Name: Sales

Add clauses:

Condition	Value	Output
is greater than	8000	Excellent
is greater than	5000	Good
Else	—	Average

Result Example

Sales	Category
9000	Excellent
5600	Good

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Sales	Category
3000	Average

Conditional Column Operators

Comparison Operators

- equals
- does not equal
- is greater than
- is greater than or equal to
- is less than
- is less than or equal to

Text Operators

- begins with
- ends with
- contains
- does not contain

Date Operators

- is after
- is before
- is on or after
- is on or before

These help when creating conditions based on:

- Categories
- Dates
- Numeric values
- Substrings

Conditional Column Examples (Practical Use Cases)

a) Flagging Missing Values

If CustomerName is null → "Missing"
Else → "Valid"

b) Profit or Loss

If Profit >= 0 → "Profit"
Else → "Loss"

c) Age Grouping

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If Age < 18 → Child
If Age < 60 → Adult
Else → Senior

d) Date-Based Classification

If OrderDate is before 01-01-2024 → “Old Orders”

e) Grading Example

Marks: <40 = Fail, 40–60 = Pass, >60 = Merit

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Removing Blank / Null Values

Nulls and blanks are extremely common in imported datasets (Excel, CSV, SQL).
They create problems such as:

- Wrong totals (SUM ignores nulls but COUNT includes blanks)
- Relationship issues in data modeling
- Errors in DAX calculations
- Incorrect visuals (empty bars or missing categories)
- Performance slowdown during refresh

Power Query offers multiple ways to remove or handle them depending on the situation.

1. Remove Blank Rows (Full Row Null or Empty)

- When the entire row is empty
- When the row has no value in any column
- When blank rows appear due to formatting in Excel files

Steps

Home → **Remove Rows → Remove Blank Rows**

- Power Query checks each row.
- If **every column** in that row = null or " " → row gets deleted.

Example

Name	Age	Salary
Riya	26	30000
Amit	32	50000

After removal:

Name	Age	Salary
Riya	26	30000
Amit	32	50000

2. Remove Nulls from a Specific Column

When to Use

- When only **specific columns** cannot contain nulls
- For key columns like ID, ProductName, Date
- When dealing with relationship keys (Primary/Foreign Key)

Method 1 → Filter Method

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1. Click the **filter dropdown** in the column
2. Uncheck **(null)**
3. OK

Method 2 → Remove Blank Values Command

Home → Remove Rows → **Remove Blank Rows**
(This works only if the entire row is blank — not column-wise)

Example

Column "Employee_ID" contains:

Employee_ID
102
null
103

After filtering null:

Employee_ID
102
103

3. Replace Null Values (Using Replace Values)

- When you cannot remove rows
- When null needs a meaningful default value
- To avoid DAX errors (division by zero, type mismatch)
- To standardize data (e.g., replace null salaries with 0)

Steps

Home → **Replace Values**

Enter:

- Value to Find → **null**
- Replace With → **desired value**

Common Use Cases

Situation	Replacement
Missing numeric value	0
Missing category	"Unknown" or "Not Provided"
Missing date	Default date (like 01-Jan-2020)

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Situation	Replacement
Missing Boolean	False

Example

Salary column:

Salary
40000
null
50000

Replace null → 0:

Salary
40000
0
50000

Filter Out Nulls (Using Filters)

- When you want to **keep rest of the rows**, excluding the null ones
- Fastest method to remove unwanted null values
- When you want to quickly clean only one column

Steps

1. Select the **column filter** dropdown
2. Uncheck **(null)**
3. Apply filter

Example

“Department” column:

Department
HR
null
IT
Finance

After filter:

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Department
HR
IT
Finance

Best For

- Removing null category
- Cleaning duplicates with missing values
- Fixing categorical data before visuals

Remove Rows with Any Nulls Across Multiple Columns

Sometimes you need to remove rows where **any** important field is null.

Steps

1. Select multiple columns
2. Right-click → **Remove Empty**
   OR
   Use Custom Filtering

When Useful

- Data quality enforcement
- Ensuring complete records for modeling
- Preparing training data for ML models

6. Handling Nulls in Numeric Columns

Options

Method	Best Use
Replace with 0	Totals, calculations, missing numeric values
Remove	If the record is incomplete
Replace with Average/Median	Data normalization
Replace with Previous/Next value	Time-series data

Handling Nulls in Text Columns

Options

Method	Best Use
Replace with “Unknown”	Category fields

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Method	Best Use
Replace with “N/A”	Missing description
Remove	Duplicate or incomplete entries

Example:
“City” column: null → “Unknown”

Why Removing/Handling Null Values Matters

1. Prevents Wrong Calculations

- SUM results less
- COUNT becomes more
- Averages become inaccurate

2. Prevents Relationship Errors

Power BI relationships break when:

- Key column has null
- Foreign key has null

3. Improves Model Quality

Clean data = faster refresh, accurate measures.

4. Avoids Visual Problems

- Empty bars
- Missing categories
- Wrong filtering

5. Essential for Data Modeling

Null values reduce data integrity and cause misinterpretation.

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Data Visualization & Power BI

~ Detailed Notes ~

- 1. Intro to Data Visualization
- 2. Importance of Data Visualization
- 3. Types of Visualizations
- 4. Data-Driven Decision Making
- 5. Introduction to Power BI
- 6. Components of Power BI
- 7. Installation & Interface
- 8. Business Intelligence Concepts
- 9. Data Analytics Lifecycle

1 Introduction to Data Visualization

What is Data Visualization?

Data Visualization is the graphical / pictorial representation of information & data using visual elements like charts, graphs, plots, infographics, maps and dashboards. It converts raw numbers into visual context so that patterns, trends, correlations & outliers become easy to spot at a glance.

> **Simple Definition:**
> "A picture is worth a thousand numbers!" — turning data into pictures so the human brain can understand it faster than reading rows of a table.

A Short History (how it evolved)

- **Before computers:** Hand-drawn maps & charts (eg. John Snow's Cholera map, 1854; Florence Nightingale's rose diagram)
- **1980s-90s:** Spreadsheet software (Excel) brought charts to everyday users
- **2000s:** BI tools (Tableau, QlikView) introduced interactive dashboards
- **Today:** Power BI, Tableau, Python (Matplotlib/Seaborn), R, cloud-based real-time dashboards

Why do we need it?

- Human brain processes **visuals up to 60,000x faster** than plain text/numbers
- ~90% of information transmitted to brain is visual
- Helps identify trends, correlations & patterns quickly
- Makes large & complex datasets understandable to everyone, not just analysts
- Supports storytelling with data (data storytelling)
- Improves memory retention of information compared to plain text

Key Elements of a Visualization

Visual / Encoding Elements

- Position (placement on axis)
- Colour (categories / intensity)
- Size (magnitude of value)
- Shape (differentiate categories)
- Length & angle (bars, pie slices)

Data Elements

- Categorical data (names, groups)
- Numerical data (quantities)
- Time-series data (dates/trends)
- Geographical data (location)
- Hierarchical data (parent-child)

1 Introduction to Data Visualization

Principles of Effective Visualization

(inspired by Edward Tufte, the "father of data visualization")

- **Clarity:** the message should be obvious at first glance
- **Accuracy:** never distort scale/axis to mislead viewer
- **Simplicity / Data-ink ratio:** remove unnecessary decoration ("chart junk") - every bit of ink should show data
- **Consistency:** same colour = same meaning throughout the report
- **Context:** always include titles, units, legends, and source of data
- **Right chart for right data** (covered in Topic 3)

Common Mistakes to Avoid

- ✘ Using 3D charts unnecessarily - distorts perception of values
- ✘ Truncated / non-zero baseline on bar charts - exaggerates difference
- ✘ Too many colours / categories in one chart (clutter)
- ✘ Using pie chart for too many categories (>6)
- ✘ Missing axis labels, legends or titles
- ✘ Wrong chart type for the data story (eg. line chart for non-time data)

Static vs Interactive Visualization

Static Visualization	Interactive Visualization
Fixed image (eg. chart in a PDF/report)	User can filter, drill down, hover for details
One-way communication	Two-way / exploratory
Made using Excel, PowerPoint	Made using Power BI, Tableau, D3.js

Exam Tip:

Data Visualization is *not just making charts pretty* - the real goal is to communicate insight clearly & accurately to drive action.

Good Visualization = Right Chart + Right Data + Clear Message + Honest Scale

2 Importance of Data Visualization

Why it matters so much...

1. **Faster Decision Making** - managers grasp insights in seconds instead of reading pages of raw reports
2. **Identifying Trends & Patterns** - sales growth, seasonal demand, customer behaviour cycles
3. **Spotting Errors / Outliers / Anomalies** - unusual spikes, drops or fraud instantly visible
4. **Better Communication** - easy to present to stakeholders, even non-technical audiences
5. **Simplifies Complex Data** - millions of rows shown meaningfully in one simple chart
6. **Increases Engagement** - visuals keep audience interested compared to raw tables
7. **Enables Comparison** - compare regions, time periods, products side-by-side instantly
8. **Supports Forecasting** - trend lines help predict future performance

Example:

A company with 10,000 rows of sales data — reading it in Excel = hours of effort.
Same data as a line chart = trend seen & understood in 5 seconds!

Benefits Across Different Fields

Field	Use of Visualization
Business	Sales dashboards, KPI tracking, revenue analysis
Healthcare	Patient vitals trends, disease-spread maps, hospital capacity
Government	Census data, budget allocation, election result maps
Sports	Player performance heatmaps, match statistics
Education	Student performance tracking, attendance trends

Finance	Stock price trends, portfolio risk analysis
Marketing	Campaign performance, customer funnel analysis

Impact on Business (why companies invest in it)

✔

- Reduces time spent on manual reporting
- ✔ Improves accuracy & reduces human error in interpretation
- ✔ Enables real-time monitoring of business health (live dashboards)
- ✔ Democratizes data - everyone in the org can self-serve insights
- ✔ Improves ROI tracking on marketing/business initiatives

➜ Visualization bridges the gap between Data and Decision-Making

3 Types of Visualizations

★ 1. Bar Chart / Column Chart

Used to compare quantities across categories. Vertical bars = Column chart, Horizontal bars = Bar chart. Good when category names are long (use horizontal).

★ 2. Line Chart

Best for showing trends over time (eg. monthly revenue, stock prices, temperature over a year). Can plot multiple lines to compare series.

★ 3. Pie Chart / Donut Chart

Shows proportion of a whole - use only when categories are few (2-6). Donut chart = pie chart with a hole in the middle, can show a total value in the center.

★ 4. Scatter Plot

Shows relationship / correlation between two numeric variables. Useful to spot clusters & outliers.

★ 5. Bubble Chart

Like a scatter plot but with a **third dimension** shown via bubble size (eg. Sales vs Profit, bubble size = number of units)

★ 6. Histogram

Shows distribution/frequency of a continuous numeric variable, grouped into "bins" or ranges (different from bar chart which compares categories)

★ 7. Box Plot (Box-and-Whisker)

Shows distribution using 5-number summary: Minimum, Q1, Median, Q3, Maximum - great for spotting outliers & spread

★ 8. Heat Map

Uses colour intensity to show magnitude - great for correlation matrices, website click maps, or calendar-based data (eg. GitHub contribution graph)

3 Types of Visualizations

★ 9. Map / Geo Visualization

Data plotted on geographic maps - eg. sales by state/country, filled/choropleth maps or point maps

★ 10. Tree Map

Shows hierarchical data as nested rectangles; size of rectangle = value, colour can show another dimension

★ 11. Gauge Chart

Shows a single KPI value against a target/range, like a car speedometer - commonly used for goal tracking

★ 12. Waterfall Chart

Shows how an initial value is affected by a series of positive/negative changes to reach a final value - very common for profit & loss breakdowns

★ 13. Funnel Chart

Shows stages of a process where the value decreases at each stage - eg. sales funnel (Leads → Contacted → Qualified → Closed)

★ 14. Word Cloud

Shows frequency of words in text data - size of word = frequency of appearance

★ 15. Card / KPI Visual

Displays a single important number prominently (eg. Total Revenue, Total Customers) - used at the top of dashboards

★ 16. Matrix / Table Visual

Shows detailed data in rows & columns, often with subtotals - like a interactive pivot table

How to Choose the Right Chart?

Comparison → Bar/Column Chart | Trend over time → Line Chart |
Proportion of whole → Pie/Donut Chart | Relationship → Scatter/Bubble |
Distribution → Histogram/Box Plot | Part-to-whole hierarchy → Tree Map |
Process stages → Funnel | Change breakdown → Waterfall

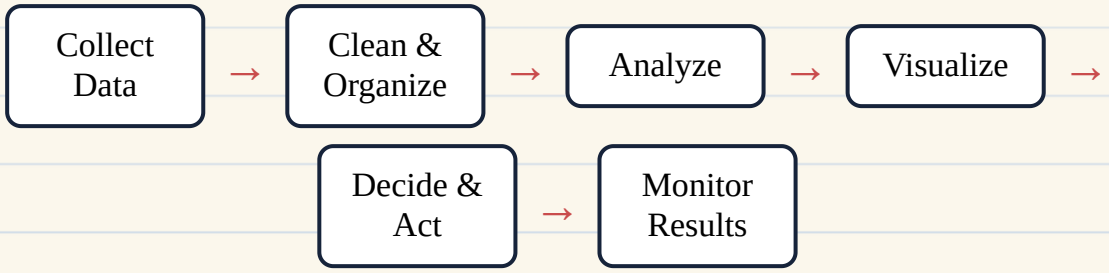
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Data-Driven Decision Making (DDDM)

Meaning

Data-Driven Decision Making means making business/organisational decisions based on actual data & analysis rather than intuition, guesswork or personal opinion alone.

Process of DDDM



Why organisations use DDDM?

- Reduces bias & guesswork in decisions
- Improves accuracy of forecasts & planning
- Increases accountability (decisions backed by evidence, not opinion)
- Helps identify new opportunities & risks early
- Improves customer understanding & personalization
- Enables continuous improvement via feedback loops

Data-Driven vs Intuition-Driven Decisions

Data-Driven	Intuition-Driven
Based on evidence/facts	Based on gut feeling/experience
Repeatable, explainable	Hard to justify to others
Slower (needs data collection)	Faster, but riskier
Reduces bias	Prone to personal bias

Challenges / Barriers to DDDM

✘

- Poor data quality (missing, duplicate, inconsistent data)
- ✘ Data silos - data scattered across different departments/systems
- ✘ Lack of data literacy among employees
- ✘ Resistance to change (people trusting "gut feeling" over data)
- ✘ Privacy & security concerns

Example

Case:

An e-commerce company notices (through dashboard) that cart abandonment is high on mobile. Data shows checkout page loads slow on mobile → Decision: optimize mobile checkout page → Result: measurable increase in conversions

Data + Insight + Action = Smart Decisions

5 Introduction to Microsoft Power BI

What is Power BI?

Power BI is a Business Intelligence & Data Visualization tool developed by Microsoft (first released 2011 as an Excel add-in, later launched as standalone product in 2015). It helps connect to various data sources, transform raw data, build interactive reports & dashboards, and share insights across an organisation.

> *In short:*
> Power BI = Data Connection + Data Transformation (Power Query) + Data Modeling (DAX) + Visualization + Sharing

Key Features

- Connects to 100+ data sources (Excel, SQL Server, Web, SharePoint, Cloud services like Azure, Google Analytics etc.)
- Drag & drop interface - no heavy coding needed for basic reports
- Powerful DAX (Data Analysis Expressions) language for custom calculations
- Real-time / live dashboards with scheduled data refresh
- Easy sharing & collaboration via Power BI Service
- AI-powered insights (Q&A natural language visual, Smart narratives, Key influencers visual)
- Custom visuals from marketplace (community-built chart types)
- Row-Level Security (RLS) - control what data each user can see

Power BI Editions / Licensing

Edition	Details
Power BI Desktop	Free - for building reports on your computer
Power BI Pro	Paid per user/month - sharing, collaboration
Power BI Premium	Capacity-based - larger datasets, more storage, dedicated resources

Power BI Embedded	For embedding reports into custom applications

Power BI vs Other BI Tools

Tool	Note
Power BI	Best MS-ecosystem integration, cost-effective
Tableau	Very strong visuals, considered more expensive
Qlik Sense	Strong associative data model / in-memory engine
Google Data Studio (Looker Studio)	Free, good for Google ecosystem data

6 Components of Power BI

The 3 Main Components

1) Power BI Desktop

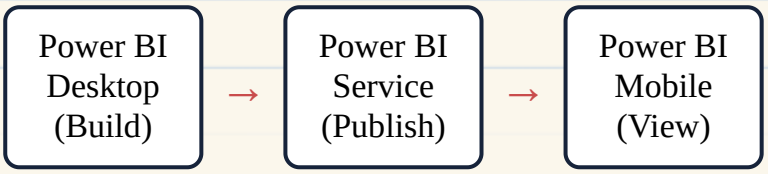
- Windows application, free to install
- Used to connect data, clean/transform (Power Query), build data model & create reports
- The "authoring tool" - where the actual report-building happens
- File saved with `.pbix` extension

2) Power BI Service (app.powerbi.com)

- Cloud-based (SaaS) platform
- Used to publish, share & collaborate on reports/dashboards
- Allows creating **Workspaces** (team areas) & **Apps** (packaged collections of reports for end users)
- Supports scheduling automatic data refresh
- Needs internet connection - accessed via web browser
- Dashboards here are made of "tiles" pinned from reports

3) Power BI Mobile

- Mobile app (Android / iOS / Windows)
- View dashboards & reports on the go, optimized layouts for phone/tablet
- Get alerts / notifications when KPIs cross a threshold
- Can annotate & share snapshot of a report



Other Related Tools / Concepts

- **Power Query** - data cleaning & transformation engine (ETL) - uses "M" language behind the scenes
- **Power Pivot / Data Model** - manages relationships between multiple tables

- **DAX (Data Analysis Expressions)** - formula language for calculated columns, measures & KPIs
- **Power BI Report Server** - on-premise (self-hosted) publishing option, for orgs that can't use cloud
- **Power BI Gateway** - a bridge connecting on-premise data sources to the cloud Power BI Service, enabling scheduled refresh
- **Dataflows** - reusable, cloud-based ETL processes shared across multiple reports
- **Workspaces** - collaborative spaces in Power BI Service to organise related reports/dashboards

7 Installation & Interface

How to Install Power BI Desktop

1. Go to official site: *powerbi.microsoft.com*
2. Click "Download Free" (Power BI Desktop)
3. Install via **Microsoft Store** (recommended - auto-updates) OR download the **.exe** installer for manual install
4. Run installer → Accept license terms → Choose install location → Install → Launch
5. Sign in with a Microsoft/Work or School account (optional for pure Desktop use, required to publish to Service)

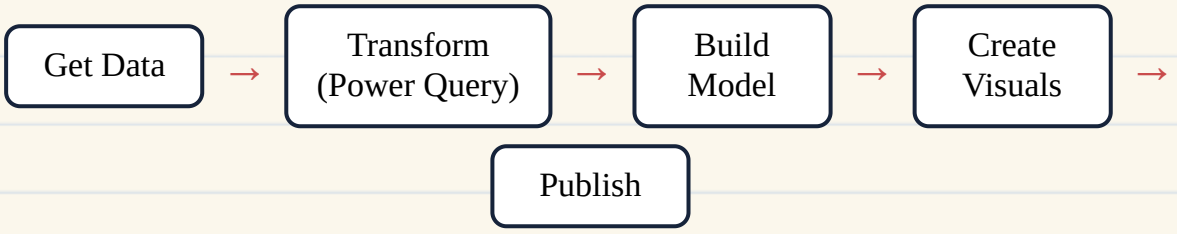
System Requirements (basic):

Windows 10/11 (64-bit), min 2 GB RAM (4 GB+ recommended), 1 GB+ disk space, .NET 4.7.2 or later, Internet Explorer 11+ (for some connectors)

First Launch - Getting Started Screen

- Option to "Get Data" immediately
- List of recent sources / pinned reports
- Link to learning resources / "What's new" changelog

Basic Workflow (first report)



7 Installation & Interface

Power BI Desktop Interface – Main Parts

1. Ribbon (Top)

Tabs: **Home** (Get Data, Transform, Refresh), **Insert** (visuals, buttons, shapes), **Modeling** (new measure/column, relationships), **View** (themes, page settings), **Optimize**, **Help**

2. Report View (Center Canvas)

The main working area where visuals/charts are placed, resized & arranged on report "pages" (tabs at the bottom)

3. Views Panel (Left side icons)

- **Report View** - build & design report pages with visuals
- **Data View** - see the raw data tables loaded into the model, row by row
- **Model View** - see & manage relationships between multiple tables (diagram view)

4. Visualizations Pane (Right)

Gallery of chart-type icons to click/insert + a Format pane below it (paint-roller icon) to customize colours, fonts, titles, data labels, axis settings etc.

5. Fields Pane (Right, below Visualizations)

Shows all tables & columns from the connected data - simply drag fields onto a visual or onto the "Fields" wells (Axis, Legend, Values) to build a chart

6. Filters Pane

Apply filters at 3 levels: **Visual level** (this chart only), **Page level** (whole page), **Report level** (entire report)

Handy Shortcuts

- `Ctrl+S` Save file

- `Ctrl+C / Ctrl+V` Copy/Paste a visual
- `Ctrl+G` Group selected visuals
- `F5` Refresh data

Workflow: Get Data → Transform (Power Query) → Build Model → Create Visuals → Format → Publish

8 Business Intelligence (BI) Concepts

What is Business Intelligence?

BI refers to the technologies, processes, strategies & tools that convert raw data into meaningful & actionable information to support better business decisions.

Core Components of BI

- **Data Warehousing** - centralized storage of data from multiple sources, optimized for analysis
- **ETL (Extract, Transform, Load)** - process to pull data from sources, clean/transform it, then load into warehouse
- **Data Mining** - discovering hidden patterns & relationships in large datasets
- **Reporting & Dashboards** - presenting data visually for consumption
- **OLAP (Online Analytical Processing)** - multidimensional analysis (slice, dice, drill-down)
- **Data Governance** - policies ensuring data quality, security & compliance

OLAP vs OLTP

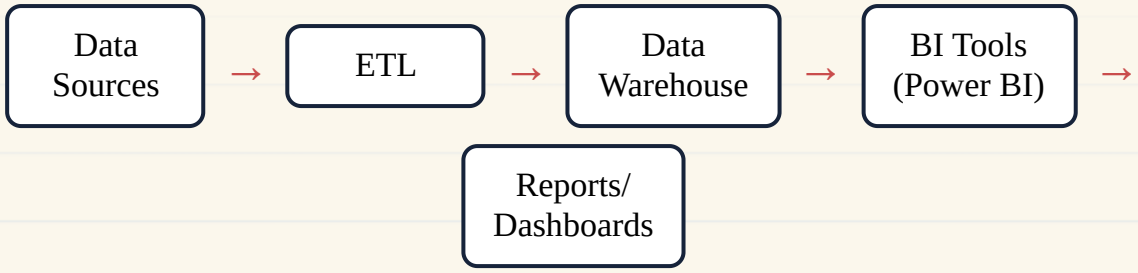
OLTP	OLAP
Online Transaction Processing	Online Analytical Processing
Day-to-day operations (eg. booking a ticket)	Analysis & reporting (eg. yearly sales trend)
Many short, simple transactions	Complex queries over large data
Normalized database design	De-normalized (star/snowflake schema)

Data Warehouse vs Data Mart

- **Data Warehouse** - centralized repository for the entire organisation's data

8 Business Intelligence (BI) Concepts

BI Architecture (simple flow)



Star Schema vs Snowflake Schema

- **Star Schema** - one central "Fact" table connected directly to multiple "Dimension" tables (simple, fast queries) - preferred in Power BI
- **Snowflake Schema** - dimension tables are further split/normalized into sub-dimensions (more complex, saves storage)

Key Terms

- **Fact Table** - holds measurable/quantitative data (eg. Sales amount, Quantity)
- **Dimension Table** - holds descriptive attributes (eg. Product, Customer, Date)
- **KPI (Key Performance Indicator)** - a measurable value showing progress towards a goal (eg. Monthly Revenue Target)
- **Dashboard** - single-page, high-level visual summary of KPIs (usually static layout, one page)
- **Report** - multi-page, detailed & interactive collection of visuals (can drill down)

Dashboard vs Report

Dashboard	Report
Single page, snapshot view	Multiple pages, detailed
Built from pinned tiles	Built directly from data model
Not very interactive/filterable	Highly interactive & filterable

Types of BI Users

User Type	Use
Executives	High level KPI dashboards for strategic decisions
Analysts	Deep-dive reports, trend & root-cause analysis
Operational Staff	Daily performance tracking & monitoring

Benefits of BI

- ✔ Improves operational efficiency & reduces manual work
- ✔ Provides competitive advantage through faster insight
- ✔ Enables faster & more accurate reporting
- ✔ Identifies new revenue opportunities & cost savings

9 Data Analytics Lifecycle

The Data Analytics Lifecycle is a step-by-step framework used to manage & execute a data analytics project effectively, from problem discovery to final result communication & deployment. (Popularized by EMC/Dell's Data Science methodology)

Step 1: Discovery

- Understand the business problem/domain
- Define project objectives & success criteria
- Identify data sources available
- Assess resources, team skills & feasibility (time, budget, tools)

Step 2: Data Preparation

- Collect/extract data from all identified sources
- Clean data - handle missing values, duplicates, wrong formats
- Explore data (EDA - Exploratory Data Analysis) to understand structure
- Set up an "analytic sandbox" - a safe workspace to experiment with data

Step 3: Model Planning

- Determine the analytical methods/techniques to use
- Explore relationships between variables (correlation)
- Choose appropriate tools (R, Python, SQL, Power BI etc.)
- Plan the overall workflow for model building

9 Data Analytics Lifecycle

Step 4: Model Building

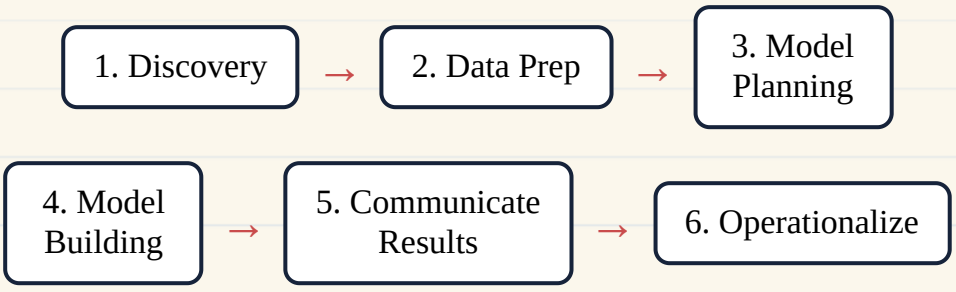
- Split data into training & testing datasets
- Build actual analytical/statistical/ML models
- Validate & tune model performance (accuracy, precision etc.)

Step 5: Communicate Results

- Compare outcomes against success criteria defined in Step 1
- Identify key findings, insights & business value delivered
- Present results to stakeholders using visualizations & simple language

Step 6: Operationalize

- Deliver final reports, dashboards, code & documentation
- Deploy model into production environment (if applicable)
- Monitor results over time & retrain/update model as needed



Roles & Tools Involved (typical)

Phase	Typical Roles	Common Tools
Discovery	Business Analyst, Data Scientist	Interviews, docs
Data Prep	Data Engineer	SQL, Power Query, Python
Model Planning/Building	Data Scientist	Python, R, ML libraries
Communicate	BI Analyst	Power BI, Tableau, PPT
Operationalize	Data/ML Engineer	Cloud platforms, APIs

Note:

This lifecycle is **iterative**, not strictly one-way! Analysts often loop back to earlier steps (eg. back to Data Prep) if results are unsatisfactory or new questions arise.

-- End of Detailed Notes -- All the best! ✍️